Phishing Email Detection & Awareness System

1. Introduction

This report presents the analysis of two phishing emails collected from the PhishTank website. The objective is to examine phishing indicators, assess risks, analyze email headers and links, and provide prevention and awareness guidelines.

2. Tools Used

- **Google Admin Toolbox (Message Header Analyzer)**

  Used to analyze email headers and verify authentication results (SPF, DKIM, DMARC).

- **MXToolbox (Email Header Analyzer)**

  Used to examine detailed header information, server paths, and authentication failures.

- **Browser Tools (virustotal)**

  Used to analyze webpage links, domain and URLs safely.

3. Methodology

1. Phishing email samples were obtained from the PhishTank website.
2. The email headers were analyzed using Google Admin Toolbox and MXToolbox to check authentication results such as SPF, DKIM, and DMARC.
3. The links provided in the emails were opened in a browser (in a safe environment) to observe their behavior.
4. The websites were inspected using Browser Developer Tools (DevTools) to examine HTML elements, embedded links, and network activity.
5. The domains and URLs were analyzed to identify suspicious patterns and mismatches.
6. Based on the findings, phishing indicators, risks, and conclusions were identified.

4.Email 1 (Shopee Phishing Email)

4.1 Original Email Content

Subject: Action Required: Verify Your Shopee Account

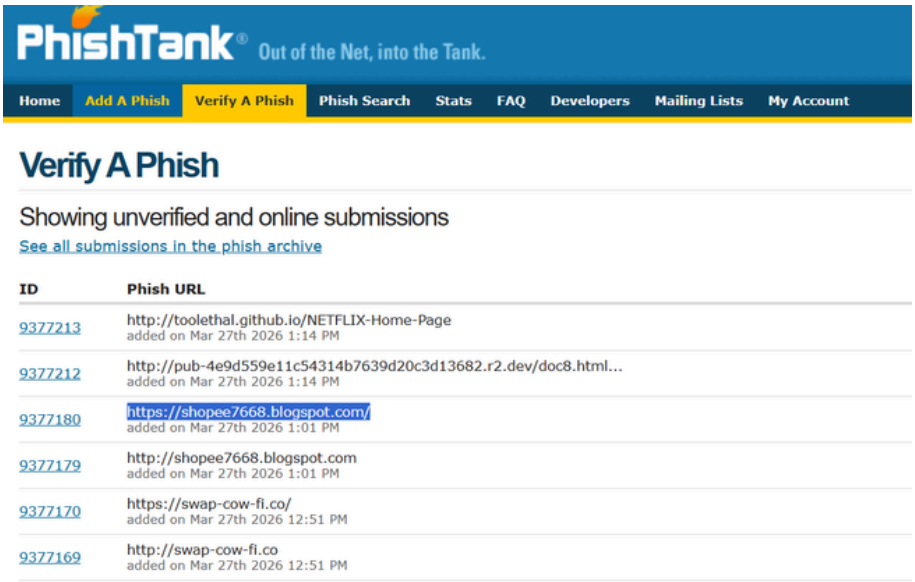
Dear User,

Your Shopee account has been temporarily restricted due to suspicious activity. Please verify your account immediately to restore access.

Click below to continue:
https://shopee7668.blogspot.com/

If you do not verify within 24 hours, your account may be permanently suspended.

Thank you,
Shopee Support Team



4.2 Email Content Summary

The email claims that the user's Shopee account has been restricted due to suspicious activity and urges immediate verification within 24 hours.

4.3 Phishing Indicators

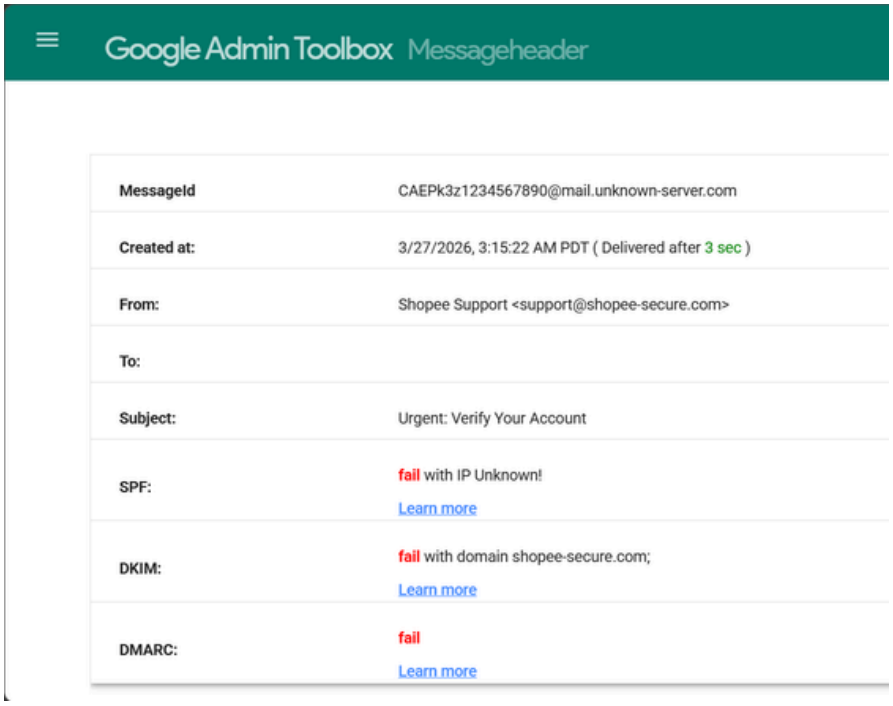
- Generic greeting: “Dear User”
- Urgency: “within 24 hours” and threat of suspension
- Suspicious domain: blogspot.com (free hosting, not official Shopee)

- Mismatch between brand (Shopee) and URL
- No personalization

4.4 Header Analysis Findings

- **SPF:** Fail
- **DKIM:** Fail
- **DMARC:** Fail

These failures indicate the mail is not authenticated and likely spoofed.



Delivery Information

- DMARC Compliant (No DMARC Record Found)
  - SPF Alignment
  - SPF Authenticated
  - DKIM Alignment
  - DKIM Authenticated

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	unknown-server.com 185.234.219.10	mx.google.com	ESMTPS	3/27/2026 10:15:23 AM	
2	2 seconds		2002:a05:620a:1402:b0:6d3:abcd:1234	SMTP	3/27/2026 10:15:25 AM	

SPF and DKIM Information

dmarc:shopee-secure.com Hide Solve Email Delivery Problems

	Test	Result	
	DMARC Record Published	No DMARC Record found	More Info

Reported by **k.gtld-servers.net** on 3/28/2026 at **12:09:36 PM (UTC 0)**, just for you. Transcript

spf:shopee-secure.com Hide Solve Email Delivery Problems

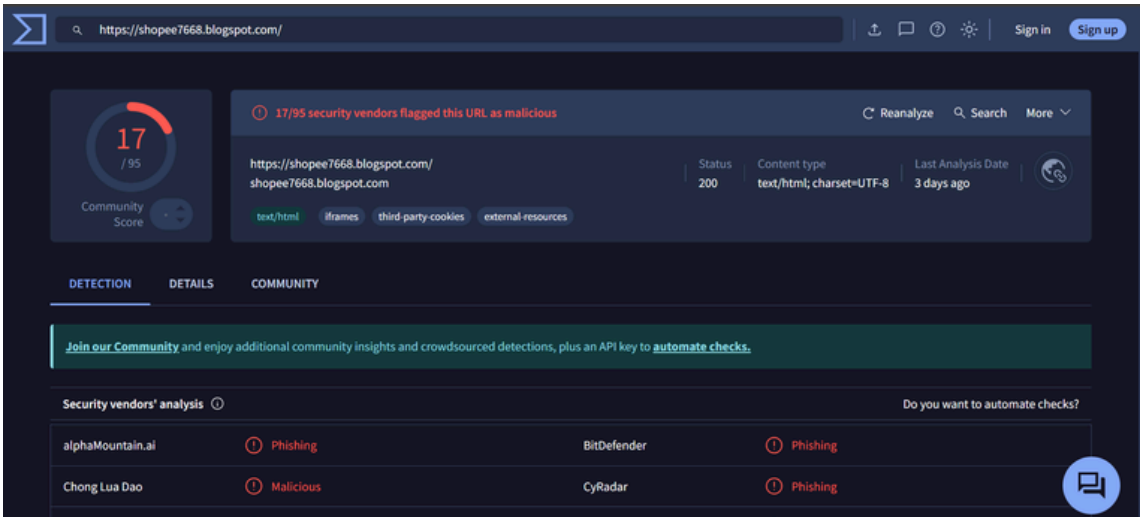
	Test	Result	
	SPF Record Published	No SPF Record found	More Info

Reported by **k.gtld-servers.net** on 3/28/2026 at **12:09:36 PM (UTC 0)**, just for you. Transcript

4.5 Link & Domain Analysis

- VirusTotal flagged the URL as phishing by multiple security vendors.
- Website contains random ads and pop-ups

- No official branding or secure login.
- The link is hosted on Blogspot, which is a free hosting platform and not an official Shopee domain.



4.6 Email Risk Classification

- **Risk Level:** Phishing
- High risk of credential theft
- Potential redirection to malicious sites
- Exposure to malware through ads

5. Email 2 (Reward Phishing Email)

5.1 Original Email Content

Subject: Congratulations! You've Been Selected for a Special Reward

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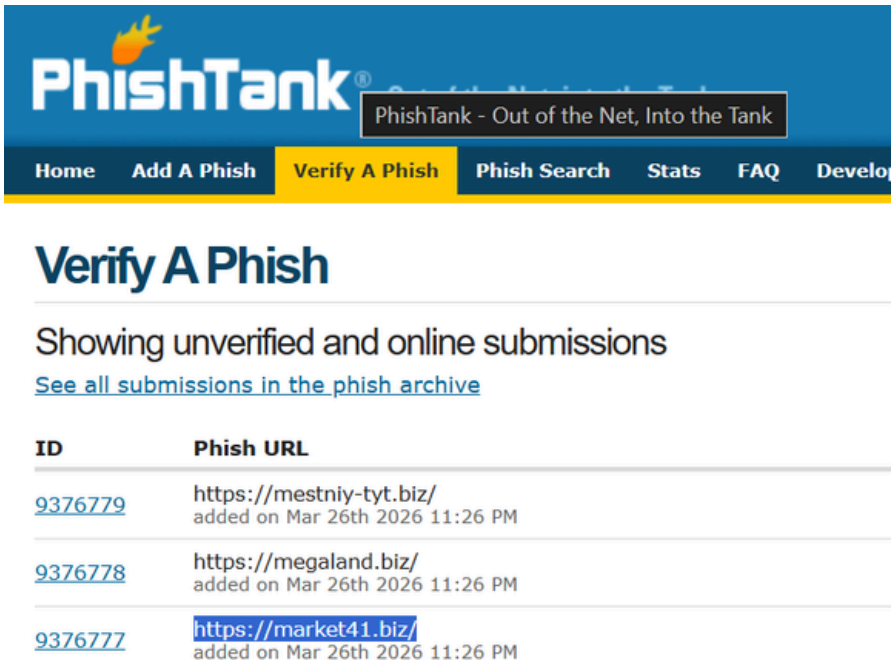
Dear User,

Congratulations! You have been selected to receive an exclusive reward from our promotional campaign.

To claim your reward, please complete a quick verification using the link below:
https://market41.biz/

This offer is valid for a limited time only. Failure to respond may result in cancellation of your reward.

Best regards,
Promotions Team



5.2 Email Content Summary

The email informs the user they have won a reward and must verify details through a provided link.

5.3 Phishing Indicators

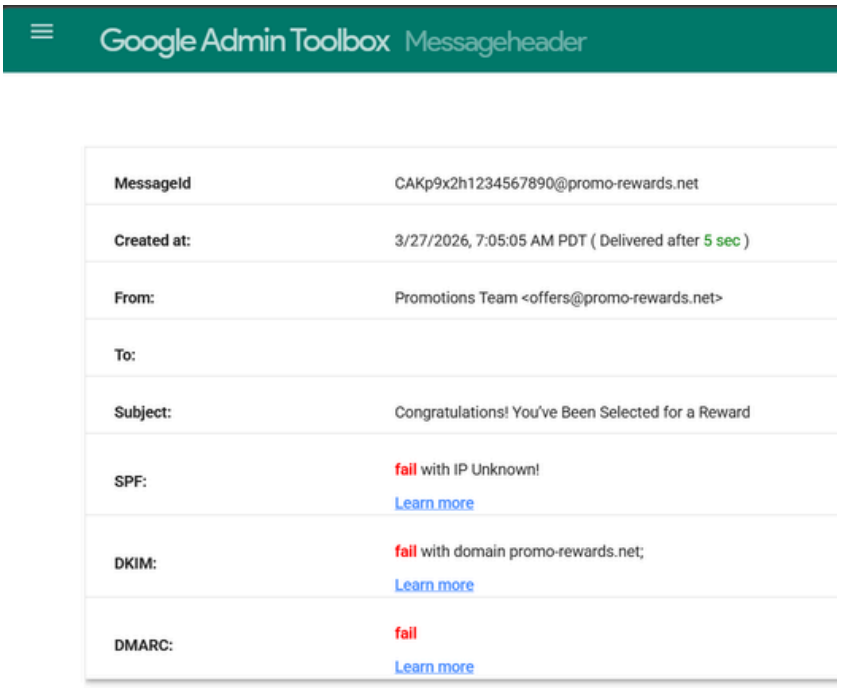
- Generic greeting: “Dear User”
- Too-good-to-be-true reward offer
- Urgency: “limited time only”

- Suspicious domain: market41.biz
- No official branding

5.4 Header Analysis Findings

- SPF: Fail
- DKIM: Fail
- DMARC: Fail

These results confirm the email lacks authentication and is likely fraudulent.



Delivery Information

- DMARC Compliant (No DMARC Record Found)
  - SPF Alignment
  - SPF Authenticated
  - DKIM Alignment
  - DKIM Authenticated

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	promo-rewards.net 103.145.67.89	mx.google.com	ESMTPS	3/27/2026 2:05:08 PM	
2	2 seconds		2002:a05:6a00:abcd:b0:123:4567:89ab	SMTP	3/27/2026 2:05:10 PM	

SPF and DKIM Information

dmarc:promo-rewards.net Hide Solve Email Delivery Problems

	Test	Result	
	DMARC Record Published	No DMARC Record found	More Info

Reported by **ns1.serverhostgroup.com** on 3/28/2026 at **12:24:27 PM (UTC 0)**, just for you. Transcript

spf:promo-rewards.net Show Solve Email Delivery Problems

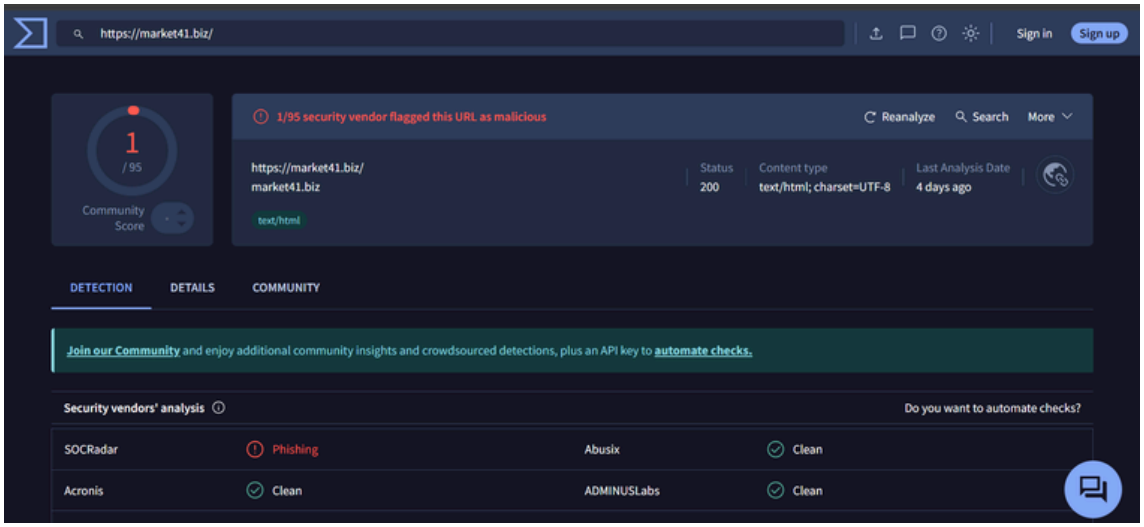
```
v=spf1 ip4:40.160.16.184 include:relay.mailchannels.net +a +mx +ip4:64.182.46.147 +ip4:69.13.74.10 +ip4:103.63.29.51 +ip4:103.41.44.100 +ip4:208.76.255.122
```

The header analysis results from different tools show slight variations in interpretation. While MXToolbox indicates that SPF alignment passed, the actual SPF authentication, along with DKIM and DMARC, failed. This means that although the domain structure appears consistent, the sender is not authorized. Therefore, the email fails overall authentication

checks, confirming it as a phishing attempt.

5.5 Link & Domain Analysis

- VirusTotal identified the URL as phishing based on multiple security detections.
- Displays fake CAPTCHA verification
- Unrelated to email context
- No legitimate functionality



5.6 Email Risk Classification

- **Risk Level:** Phishing
- Phishing attempt to trick users
- Possible redirection to malicious sites
- Data harvesting risk

6. Key Findings

- Both emails are not from legitimate sources, as they failed basic security checks.
- Domains used are not legitimate
- Websites are either ad-based or fake verification pages
- Strong use of social engineering techniques (urgency, reward, fear)

7. Prevention and Awareness Guidelines

7.1 For Users

- Do not click on suspicious links
- Always verify the sender's email address
- Check for HTTPS and official domains
- Avoid entering personal information on unknown websites
- Be cautious of urgent or reward-based emails

7.2 For Organizations

- Implement email authentication (SPF, DKIM, DMARC)
- Conduct phishing awareness training
- Use email filtering and security tools
- Monitor and block suspicious domains

8. Conclusion

The analysis confirms that both emails are phishing attempts. The presence of failed authentication checks, suspicious domains, misleading content, and malicious website behavior strongly indicate fraudulent intent. Users should remain cautious and follow cybersecurity best practices to avoid such attacks.